

markdown2html5-base

Full Markdown Functionality Reference

This reference covers all `markdown2html5-base` features.

Table of Contents

- [Programmatic Usage](#)
- [1. Headings \(H1-H6\)](#)
- [2. Inline Text Styling](#)
- [3. Links and Images](#)
- [4. Fenced Code Blocks](#)
- [5. Horizontal Rules](#)
- [6. Lists](#)
- [7. TOC \(Table of Contents\)](#)
- [8. Blockquotes](#)
- [9. Tables](#)
- [10. Definition Lists](#)
- [11. Footnotes](#)
- [12. Language Markers](#)
- [13. Ruby Annotations](#)
- [14. Emoji Shortcodes](#)
- [15. Typography and Legal Marks](#)
- [16. Hard Line Breaks](#)
- [17. HTML Comments](#)
- [18. YAML Front Matter](#)
- [19. Backslash Escaping](#)
- [20. Paragraphs](#)
- [21. CSS Styles](#)

Programmatic Usage

```
from markdown2html5_base import MarkdownToHTML
converter = MarkdownToHTML()
html = converter.convert("# Hello")
print(html)
```

1. Headings (H1-H6)

Markdown	Output HTML
# Heading 1	Heading 1
## Heading 2	Heading 2
### Heading 3	Heading 3
#### Heading 4	Heading 4
##### Heading 5	Heading 5
##### Heading 6	Heading 6

Custom ID: ## Section {#sec1} => <h2 id="sec1">Section</h2>

2. Inline Text Styling

Markdown	Output HTML
bold italic	bold italic
___bold italic___	bold italic
bold	bold
__bold__	bold
italic	italic
italic	italic
~~strikethrough~~	<s>strikethrough</s>
==highlight==	<mark>highlight</mark>
~subscript~	_{subscript}
^^underline^^	<u>underline</u>
^superscript^	^{superscript}
`code`	<code>code</code>

As in CommonMark, an underscore does not open or close emphasis inside a word, so identifiers keep their underscores: `snake_case_name` → `snake_case_name`, not `snake<em>case</em>name`. Use `*` for emphasis, or escape the underscores (`snake\_case\_name`), when the underscores must remain visible.

3. Links and Images

Markdown	Output HTML
[text](url)	<code>text</code>
[text](url "Title")	<code>text</code>
![alt](src)	<code><figure></figure></code>
![alt](src "Title")	<code><figure><figcaption>Title</figcaption></figure></code>

A link destination is taken verbatim and is never processed for emphasis or typography, so [link_text] (target_x) renders as `<a href="target_x">link_text</a>`. Balanced parentheses are kept ([text] (http://e.com/a_(b))), and a destination containing spaces can be wrapped in angle brackets ([text] (<http://e.com/a b>)).

4. Fenced Code Blocks

A language tag after the opening fence (three backticks before and after the code) is rendered as a `<div class="code-lang">/python/</div>` label above the `<code>` element:

```
<div class="code-lang">&sol;python&sol;</div><pre><code>def hello():
    print("Hi")</code></pre>
```

Without a language tag, the code renders plainly as `<pre><code>`.

5. Horizontal Rules

--- / *** / ___ (3+ chars) => `<hr>`

6. Lists

1. **Unordered:** * Item or - Item => `<ul><li>Item</li></ul>`

2. **Ordered:** 1. Item => `<ol><li>Item</li></ol>`

3. **Task:**

- [] todo => `<ul><li><input type="checkbox" disabled> todo</li></ul>`

1. [x] done => `<ol><li><input type="checkbox" checked disabled> done</li></ol>`

4. **Nested:** indent an item with 2 (or 4) spaces or a tab to nest a list one level deeper. Nesting works to any depth, and each level may switch freely between bullets and numbers:

```
- Parent
  - Child
    1. Grandchild
```

=>

```

<ul>
  <li>Parent
    <ul>
      <li>Child
        <ol>
          <li>Grandchild</li>
        </ol>
      </li>
    </ul>
  </li>
</ul>

```

7. TOC (Table of Contents)

The Table of Contents (TOC) combines the styles of Heading 2, unordered lists, and links.

```
## Table of Contents {#toc}
```

- [1. Headings (H1-H6)](#1-headings-h1-h6)
- [2. Inline Text Styling](#2-inline-text-styling)
- [3. Links and Images](#3-links-and-images)

=>

```

<h2 id="toc">Table of Contents</h2>
<ul>
  <li><a href="#1-headings-h1-h6">1. Headings (H1-H6)</a></li>
  <li><a href="#2-inline-text-styling">2. Inline Text Styling</a></li>
  <li><a href="#3-links-and-images">3. Links and Images</a></li>
</ul>

```

8. Blockquotes

> text => <blockquote><p>text</p></blockquote> : blank lines within a blockquote split into separate <p> tags.

9. Tables

Supports <thead> , <tbody> , <tfoot> , and alignment:

Separator	Alignment
:---	left
:---:	center
---:	right

Footer: a row of `=` signs below the separator columns, after the body rows, renders a `<tfoot>` .

	Product		Qty		Price	
	:-----		:--:		----:	
	Apples		2		\$3.00	
	Bananas		3		\$1.50	
	Cherries		1		\$4.00	
	=====		=====		=====	
	Total		6		\$8.50	

=>

```
<table>
  <thead>
    <tr>
      <th style="text-align:left;">Product</th>
      <th style="text-align:center;">Qty</th>
      <th style="text-align:right;">Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td style="text-align:left;">Apples</td>
      <td style="text-align:center;">2</td>
      <td style="text-align:right;">$3.00</td>
    </tr>
    <tr>
      <td style="text-align:left;">Bananas</td>
      <td style="text-align:center;">3</td>
      <td style="text-align:right;">$1.50</td>
    </tr>
    <tr>
      <td style="text-align:left;">Cherries</td>
      <td style="text-align:center;">1</td>
      <td style="text-align:right;">$4.00</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td style="text-align:left;">Total</td>
      <td style="text-align:center;">6</td>
      <td style="text-align:right;">$8.50</td>
    </tr>
  </tfoot>
</table>
```

10. Definition Lists

```
Term
: Definition 1
: Definition 2
```

=>

```
<dl>
  <dt>Term</dt>
  <dd>Definition 1</dd>
  <dd>Definition 2</dd>
</dl>
```

11. Footnotes

Reference: `[^1]` => `<sup id="fnref:1"><a href="#fn:1" class="footnote-ref">1</a></sup>`

Definition: `[^1]: Text` at bottom => rendered in `<div class="footnotes"><ol>...</ol></div>`

Identifiers may contain letters, digits, underscores and dashes, e.g. `[^see_note_1]`.

12. Language Markers

Annotate blocks or inline text with a language using a valid BCP 47 tag (e.g. `de`, `fr`, `zh-Hans`).

- **Block level:** Place `{:lang}` (with a space after it) at the very start of a line, before the Markdown tag. It renders as the global `lang` attribute on the resulting element and needs no closing marker:

```
{:de} # Überschrift
{:fr} Un paragraphe français.
{:ru} - Пункт списка
{:de} > Ein Zitat
{:de} | Kopf   | Kopf   |
      | -----| -----|
      | Zelle  | Zelle  |
```

=>

```
<h1 lang="de">Überschrift</h1>
<p lang="fr">Un paragraphe français.</p>
<ul lang="ru">
  <li>Пункт списка</li>
</ul>
<blockquote lang="de">...</blockquote>
<table lang="de">...</table>
```

- **Inline level:** Wrap text with `{:lang}...{:}` to render a `<span lang="...">`:

A French phrase `{:fr}"L'État c'est moi"{:}` is traditionally attributed to King Louis XIV of France.

=>

`<p>A French phrase <span lang="fr">"L'État c'est moi"</span> is traditionally attributed to King Louis XIV of France</p>`

13. Ruby Annotations

`{日本語|にほんご} => <ruby>日本語<rp>(</rp><rt>にほんご</rt><rp>)</rp></ruby>`

14. Emoji Shortcodes

- `:joy:` 😄
- `:smile:` 😊
- `:heart:` ❤️
- `:thumbsup:` 👍
- `:thumbsdown:` 👎
- `:wink:` 😉
- `:tada:` 🎉
- `:rocket:` 🚀
- `:fire:` 🔥
- `:star:` ★
- `:cry:` 😭
- `:thinking:` 🤔
- `:100:` 100
- `:sparkles:` ✨
- `:eyes:` 👁️
- `:bulb:` 💡
- `:warning:` ⚠️
- `:ok:` 🏆
- `:check_mark:` ✓

15. Typography and Legal Marks

Name	Input	HTML Output
Copyright	(c)	©
Trademark	(tm)	™
Registered Trademark	(r)	®
Plus-Minus	+/-	±
Not Equal To	!=	≠
Logical Equivalence	<=>	⇔
Less-Than or Equal To	<=	≤
Greater-Than or Equal To	>=	≥
Right Arrow	->	→
Left Arrow	<-	←
Up Arrow	↑	↑
Down Arrow	↓	↓
Logical Implication	=>	⇒
Solidus (Slash)	/	/
Reverse Solidus (Backslash)	\	\
One-Half	1/2	½
One-Third	1/3	⅓
Two-Thirds	2/3	⅔
One-Quarter	1/4	¼
Three-Quarters	3/4	¾
Left Angle Quote	<<	«
Right Angle Quote	>>	»
Left Double Quote	“	“
Right Double Quote	”	”
Smart Double Quotes	"text"	“text”
Smart Single Quotes	'text'	‘text’
Straight Apostrophe	'	'
Em Dash	---	—

Name	Input	HTML Output
En Dash	--	–
Ellipsis	...	…
Non-Breaking Space	 	

16. Hard Line Breaks

End line with two spaces or backslash: `<br />`

17. HTML Comments

[comment]: # => `<!--comment-->`

18. YAML Front Matter

If the file begins with a YAML front matter block between `---` lines, the converter emits a complete HTML5 document with the metadata in `<head>` and the body content between `<body>` tags. Without front matter, the output stays a bare HTML fragment. The `--css` option (or `include_css=True`) embeds the default `<style>` block regardless of whether front matter is present.

```
---
lang: en
title: My Document
author: Jane Doe
description: A short description.
keywords: python, markdown, html5
published: 2026-08-09
---
```

- `lang` becomes the `<html lang="...">` attribute (a valid BCP 47 tag).
- `title` becomes the `<title>` element.
- `author`, `description`, `keywords`, and `published` become `<meta name="..." content="..." />` tags.
- A default `<style>` block with viewing-friendly CSS is embedded in `<head>`.

```

<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="author" content="Jane Doe" />
    <meta name="description" content="A short description." />
    <meta name="keywords" content="python, markdown, html5" />
    <title>My Document</title>
    <meta name="published" content="2026-08-09" />
    <style>
      /* default viewing-friendly CSS */
    </style>
  </head>
  <body>
    ...
  </body>
</html>

```

Any other keys are ignored, and if the file has no front matter at all, the output remains a bare fragment (unless `--css / include_css=True` is used, in which case it becomes a full document).

19. Backslash Escaping

Escape any special char: `\*`, `\#`, `\[`, etc.

Escapable: `\`*_{}[]()#+-.!|~^=:<>`

20. Paragraphs

Consecutive text lines merge into `<p>`. Blank lines separate paragraphs.

Empty line after list close => `<!-- -->` preserves whitespace.

21. CSS Styles

The converter (with `--css` option or `include_css=True`) embeds a default `<style>` block in `<head>` that provides viewing-friendly styling, regardless of YAML front matter. The predefined CSS rules include:

```
body {
  padding: 20px;
  font-family: "Noto Serif", "Liberation Serif", "Times New Roman", Times, serif;
  font-size: 18px;
  line-height: 1.4;
  color: #000000;
  background-color: #ffffff;
}
h1 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 32px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h2 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 28px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h2#toc { font-style: italic; }
h3 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 24px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h4 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 20px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
```

```
}  
h5 {  
  margin-top: 1.2em;  
  margin-bottom: 0.6em;  
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;  
  font-weight: bold;  
  font-size: 18px;  
  hyphens: auto;  
  word-break: normal;  
  overflow-wrap: break-word;  
  text-wrap: balance;  
}  
h6 {  
  margin-top: 1.2em;  
  margin-bottom: 0.6em;  
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;  
  font-weight: bold;  
  font-size: 18px;  
  font-style: italic;  
  hyphens: auto;  
  word-break: normal;  
  overflow-wrap: break-word;  
  text-wrap: balance;  
}  
p {  
  hyphens: auto;  
  hyphenate-limit-chars: 6 3 3;  
  word-break: normal;  
  overflow-wrap: break-word;  
}  
hr {  
  height: 4px;  
  margin-top: 32pt;  
  border: none;  
  background-color: #000000;  
}  
blockquote {  
  margin-left: 0;  
  padding-left: 20px;  
  border-left: 8px solid #f5f5f5;  
  hyphens: auto;  
  hyphenate-limit-chars: 6 3 3;  
  word-break: normal;  
  overflow-wrap: break-word;  
}  
mark {  
  padding: 0 2px;  
  border-radius: 4px;  
  background-color: #ffff00;  
  color: #000000;  
}  
a:link { color: #0000cd; }  
a:visited { color: #9400d3; }
```

```
a:hover { outline: none; color: #000080; }
a:focus { outline: none; color: #000080; }
a:active { color: #dc143c; }
ol {
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ul {
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
li {
  position: relative;
  padding-left: 20px;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ol li input[type="checkbox"], ul li input[type="checkbox"] {
  vertical-align: baseline;
  accent-color: #000000;
  opacity: 1;
  cursor: not-allowed;
}
dt {
  font-weight: bold;
  hyphens: auto;
  word-break: break-word;
  overflow-wrap: anywhere;
}
dd {
  position: relative;
  margin-left: 0;
  padding-left: 20px;
  font-style: italic;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
code {
  padding: 2px 4px;
  border-radius: 4px;
  font-family: "Noto Sans Mono", "Liberation Mono", "Courier New", Courier, monospace;
  font-size: 0.9em;
  line-height: 1;
  hyphens: none !important;
  white-space: normal;
```

```

word-break: break-all;
overflow-wrap: anywhere;
}
pre {
  max-width: 100%;
  margin: 0;
  padding: 20px;
  border: 1px solid #000000;
  background-color: #f5f5f5;
  overflow: auto;
  scrollbar-color: #000000 transparent;
  white-space: pre-wrap;
  word-break: break-all;
  overflow-wrap: anywhere;
}
pre > code {
  display: block;
  margin: 0;
  padding: 0;
  border: none;
  border-radius: 0;
  line-height: 1.2;
  background-color: transparent;
  overflow: visible;
  hyphens: none !important;
  white-space: pre;
  word-break: normal;
  overflow-wrap: normal;
}
div.code-lang {
  display: block;
  padding: 10px 20px;
  font-family: "Noto Sans Mono", "Liberation Mono", "Courier New", Courier, monospace;
  font-size: 0.9em;
  line-height: 1;
  background-color: #000000;
  color: #ffffff;
  font-weight: bold;
}
table {
  margin-top: 32pt;
  border-collapse: collapse;
}
th { padding: 10px 12px; border: 1px solid #000000; font-weight: bold; }
td { padding: 10px 12px; border: 1px solid #000000; }
thead tr { background-color: #000000; color: #ffffff; }
thead th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
thead td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tbody th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tbody td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tfoot tr { background-color: #f5f5f5; font-style: italic; }
tfoot th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tfoot td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }

```

```

figure {
  display: block;
  margin: 0;
}
figure img {
  display: block;
  max-width: 100%;
  height: auto;
}
figcaption {
  text-align: left;
  font-style: italic;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ruby { ruby-position: over; ruby-align: space-around; }
rt {
  letter-spacing: 0.05em;
  font-size: 0.55em;
  line-break: strict;
  white-space: nowrap;
  overflow-wrap: normal;
}
rp { display: none; }
span[lang="ja"] {
  font-family: "Noto Serif CJK JP", "Source Han Serif JP", "源ノ明朝", "Source Han Serif", "Hiragino Mincho ProN", "Hiragino Mincho Pro", "IPAexMincho", "IPAMincho", "MS PMincho", "MS Mincho", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-CN"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hans"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-TW"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU", "MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hant"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU", "MingLiU", serif;

```

```
word-break: break-all;
line-break: normal;
}
span[lang="zh-HK"] {
  font-family: "Noto Serif CJK HK", "Source Han Serif HK", "思源宋體 香港", "思源宋體", "Source Han
Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "MingLiU_HKSCS", "PMingLiU", "MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="ko"] {
  font-family: "Noto Serif CJK KR", "Source Han Serif KR", "본명조", "Source Han Serif",
"AppleMyungjo", "UnBatang", "은바탕", "Batang", serif;
  word-break: break-all;
  line-break: normal;
}
```